Microsoft 365

ICE-3207 Software and Requirement Engineering

Nazneen Akhter

Assistant Professor

Department of ICT, FST

Answer

- How many of you have developed a software project?
- Why do software projects fail?
- Is coding the most difficult part of software development?

Example





"Can a single programmer build these systems?"





What is Software Engineering

Definition by IEEE: Software Engineering is the systematic, disciplined, and measurable approach to developing, operating, maintaining, and evolving software.

Programming ≠ Software Engineering

Software = Code + Documentation + Testing + Maintenance + Teamwork

Why Software Projects Fail

Common reasons:

- Poor requirements
- Lack of planning
- Budget overrun
- Missed deadlines
- Poor communication
- Inadequate testing
- Frequent requirement changes

Example:

Imagine your university asks you to build an online course registration system. If you start coding without asking what the registrar, students, and teachers need, the project is likely to fail.

Software Development Life Cycle

Planning
↓
Requirement Analysis
↓
Design
↓
Implementation
↓
Testing
↓
Deployment
↓
Maintenance
↺



Practice your presentation

1. Planning
- What problem are we solving?
- Cost estimation
- Feasibility

2. Requirements
- Functional requirements
- Non-functional requirements

3. Design
- Database
- UI
- Architecture

6. Implementation
- Coding
- Git
- Team collaboration

5. Testing
- Find bugs
- Verify requirements

4. Deployment
- Release to users

7. Maintenance
- Bug fixes
- New features
- Performance improvements

Example: Online Library System

SDLC Phase	Example Activity		Common Industry Tools
Planning	Define project scope, timeline, feasibility, assign tasks	Need a digital library	Jira, Trello, Asana, Microsoft Project, Notion
Requirements Analysis	Gather and document functional & non-functional requirements	Search books, borrow books, return books	Microsoft Word, Confluence, Google Docs, Jira, Miro
Design	Create UI, database schema, system architecture, UML diagrams	Database, UI	Figma, Draw.io, Lucidchart, StarUML, Visual Paradigm, MySQL Workbench
Implementation	Develop frontend, backend, database	Develop the system	Visual Studio Code, IntelliJ IDEA, Eclipse, Visual Studio, Git, GitHub/GitLab
Testing	Unit, integration, system, and API testing	Check borrowing and returning	JUnit, pytest, Selenium, Postman, Cypress, JMeter
Deployment	Deploy the application to production	Launch for students	Docker, GitHub Actions, Jenkins, Nginx, AWS, Microsoft Azure, Google Cloud
Maintenance	Fix bugs, monitor performance, release updates	Add online payment, improve performance	GitHub Issues, Jira, Sentry, Grafana, Prometheus, Datadog